

## Answers

### VARC

|              |              |              |              |                 |                 |              |                 |
|--------------|--------------|--------------|--------------|-----------------|-----------------|--------------|-----------------|
| 1. <b>B</b>  | 2. <b>D</b>  | 3. <b>C</b>  | 4. <b>B</b>  | 5. <b>B</b>     | 6. <b>C</b>     | 7. <b>C</b>  | 8. <b>D</b>     |
| 9. <b>A</b>  | 10. <b>A</b> | 11. <b>B</b> | 12. <b>B</b> | 13. <b>D</b>    | 14. <b>B</b>    | 15. <b>C</b> | 16. <b>D</b>    |
| 17. <b>D</b> | 18. <b>C</b> | 19. <b>B</b> | 20. <b>C</b> | 21. <b>2143</b> | 22. <b>3214</b> | 23. <b>D</b> | 24. <b>2431</b> |

### LRDI

|              |              |              |               |              |               |              |              |
|--------------|--------------|--------------|---------------|--------------|---------------|--------------|--------------|
| 25. <b>C</b> | 26. <b>D</b> | 27. <b>C</b> | 28. <b>A</b>  | 29. <b>B</b> | 30. <b>50</b> | 31. <b>C</b> | 32. <b>B</b> |
| 33. <b>A</b> | 34. <b>C</b> | 35. <b>2</b> | 36. <b>A</b>  | 37. <b>3</b> | 38. <b>3</b>  | 39. <b>D</b> | 40. <b>A</b> |
| 41. <b>A</b> | 42. <b>A</b> | 43. <b>8</b> | 44. <b>48</b> |              |               |              |              |

### Quant

|                  |               |               |                |                |               |              |               |
|------------------|---------------|---------------|----------------|----------------|---------------|--------------|---------------|
| 45. <b>43200</b> | 46. <b>C</b>  | 47. <b>C</b>  | 48. <b>C</b>   | 49. <b>B</b>   | 50. <b>66</b> | 51. <b>A</b> | 52. <b>82</b> |
| 53. <b>C</b>     | 54. <b>84</b> | 55. <b>A</b>  | 56. <b>160</b> | 57. <b>B</b>   | 58. <b>C</b>  | 59. <b>D</b> | 60. <b>C</b>  |
| 61. <b>A</b>     | 62. <b>D</b>  | 63. <b>34</b> | 64. <b>A</b>   | 65. <b>111</b> | 66. <b>44</b> |              |               |

## Explanations

### VARC

#### 1. **B**

Option A: This is a possible implication because the passage states that through meditation and the right attitude, one allows emotions to happen to themselves, which implies a passive reception of those experiences.

Option B: The passage does not state or imply that meditation allows for certain out-of-body experiences or that it allows an individual to gain distance from their emotions in that way. The passage specifically highlights that through meditation and the right attitude, one allows emotions to happen to themselves and then observes those emotions without necessarily acting on them, which allows one to achieve some distance and decide what has value and what does not have value. It does not mention anything about out-of-body experiences or any other kind of distance that is not achieved through the act of observing emotions.

Option C: This is a possible implication because the passage states that the second movement is what one does next about the emotion and that it occurs after thinking and is under one's control. Observing emotions in a distant manner, as described in the quote, would involve thinking and control and would therefore correspond to the second movement referred to earlier.

Option D: This is a possible implication because the quote mentions that observing emotions allows an individual to decide what has value and what does not have value, implying that emotional responses can make this distinction difficult.

Hence, Option B is the correct answer.

VIDEO SOLUTION

2. D

Based on the discussion, Option D is the correct answer: the passage describes an example of what might have happened if an army general accused Marcus Aurelius of treason in front of other officers, implying that Marcus Aurelius was a leader in the Roman army. The other options are neither mentioned nor implied in the passage and are therefore not supported by the information provided.

Option A: is incorrect because the passage does not mention anything about Marcus Aurelius feeling humiliated or embarrassed by the accusation; it only describes the immediate feeling and other reactions that may have occurred in response to the stimulus of the accusation, such as surprise and anger.

Option B: is incorrect since the author does not label Marcus Aurelius as a Stoic or associate him with the philosophy of Stoicism in any way; he only uses him as an example of what might have happened in a specific situation involving an army general accusing him of treason.

Option C: is incorrect since the passage does not state that Marcus Aurelius was plotting revenge or seeking justice; it only mentions that one of the potential second movements that Marcus Aurelius might have chosen in response to the accusation could have been a plot to seek revenge. However, it does not state that this is what actually happened or that it was the only possible second movement that Marcus Aurelius could have chosen.

VIDEO SOLUTION

3. C

In this question, we need to find a statement which is in line with the ideas given in the passage(then, if it is false, it will contradict the passage).

Option A: "Through meditation and the right attitude, one allows emotions to happen to oneself (it is impossible to prevent this), but one is advised to observe the emotions without necessarily acting on them; one achieves some distance and decides what has value and what does not have value."

The above excerpt was written to describe the similarities between Stoicism and Buddhism. Since option A is not in line with the above excerpt's idea, it is not the correct option.

Option B: This option can easily be refuted on the basis of the information given in the first line of the passage. It cannot be inferred that Zeno survived into the Roman era until about AD 300.

Option C: "The Epicureans, first associated with the Greek philosopher Epicurus . . . held a similar view, believing that people should enjoy simple pleasures, such as good conversation, friendship, food, and wine, **but not be indulgent in these pursuits and not follow passion for those things that hold no real value like power and money.**"

Since this option reiterates the idea mentioned in the underlined portion of the above excerpt, this is the correct option.

Thus, the correct option is C.

VIDEO SOLUTION

4. B

The passage states that "the Stoic idea of developing virtue in oneself, of becoming a good person, which the Stoics believed we could do because we have a touch of the divine, laid the foundation for the three monotheistic religions: Judaism, Christianity, and Islam." This aligns with the claim in Option B. Options A and C are incorrect as the passage states that the Stoics believed in freeing oneself of nearly all desires, including excessive desires for money, power, and sexual gratification [A], and that the Epicureans believed in enjoying simple pleasures but not being indulgent or pursuing things with no real value [not sufficient to validate C]. Similarly, we cannot substantiate the statement in D.

Hence, Option B is the correct answer.

VIDEO SOLUTION

5. B

The passage discusses cultural differences in the concept of a copy and the value placed on originality, particularly in relation to art and religious buildings. In China, copies (fuzhipin) are considered to be of equal value to the original and do not carry negative connotations, while in the Western world, the idea of an unassailable original has historically held more importance. This difference in perspective has led to misunderstandings and tensions between China and Western museums when Chinese museums send copies abroad.

Based on the above, a Chinese museum would be unlikely to consider Option B [Pablo Picasso's painting of Vincent van Gogh's original painting, identical in every respect] as having less value than the original. This is because the Chinese concept of a copy (fuzhipin) refers to exact reproductions of the original that are considered to be of equal value to the original and do not carry negative connotations. Contrarily, Option A - a painting of Vincent van Gogh's original painting by Pablo Picasso with Picasso's signature - would not be considered a fuzhipin as it is not an exact reproduction of the original and bears the signature of a different artist. Similarly, Options C and D would also not be considered a fuzhipin since they are not an exact reproduction of the original [but merely different versions/formats].

Hence, Option B is the correct choice.

VIDEO SOLUTION



6. C

{Back in the 17th century [in the West], excavated artworks from antiquity were treated quite differently from today. They were not restored in a way that was faithful to the original. Instead, there was massive intervention in these works, changing their appearance. .}

Based on the passage, the scenario in Option C [A 17th-century French artist who adhered to a Christian worldview would need to be completely true to the original intent of a painting when restoring it] is unlikely to follow from the arguments in the passage. The passage mentions that in the 17th century, excavated artworks from antiquity were treated differently from how they are today and were not restored in a way that was faithful to the original. Instead, there was "massive intervention" in these works, changing their appearance. This suggests that the idea of an unassailable original may not have held as much importance in the 17th century as it does today. Therefore, it is unlikely that a 17th-century French artist who adhered to a Christian worldview would necessarily need to be completely true to the original intent of a painting when restoring it. Contrarily, we cannot definitively comment on the other scenarios - A, B, and D.

Hence, Option C is the correct choice.

VIDEO SOLUTION

7. C

{The Ise Grand Shrine [in Japan] is 1,300 years old for the millions of Japanese people who go there on pilgrimage every year. But in reality this temple complex is completely rebuilt from scratch every 20 years. . . }

While we know that the cathedral of Freiburg Minster is continually undergoing restoration, the same cannot be said about Ise Grand Shrine - we are told that it is rebuilt periodically, but there is no information not substantiate that it is being continually restored [at least in the same sense as that conveyed in the passage]. The idea of restoration is based on how these monuments are being rebuilt.

Hence, Option C is the correct choice.

VIDEO SOLUTION

8. D

Option A: The value placed on an unassailable original in the Western world may discourage the simultaneous display of multiple copies of a painting, as the original is considered more valuable and authentic. Hence, Option A is valid.

Option B: This is a valid option because the value placed on the original artwork in the Western world may lead to the regular employment of craftsmen who are responsible for preserving and restoring original works of art. This can include tasks such as examining the artwork for damage and replacing eroded or damaged materials [restoration].

Option C: It is true that the focus on the original in the Western world may discourage interventions in ancient art that would alter the appearance of the original. In the past, ancient artworks were frequently altered during restoration, but this practice has become less common in recent times as the value of preserving the original appearance of the artwork has increased. Thus, Option C is plausible.

Option D: The passage discusses how the idea of an original work of art that cannot be altered developed in the Western world and how this intellectual position has led to different attitudes towards cloning between Europe and Asia. However, the passage does not directly mention that the value placed on an unassailable original has discouraged or influenced attitudes towards human cloning in the Western world.

Hence, Option D is the correct answer.

VIDEO SOLUTION

9. A

Option A best expresses the claim made in the statement - "*In order to confirm their abnormal status, many of the Undead were often accorded attributes, which defied the natural order of things. . .*"

The passage states that in order to confirm their abnormal status, the Undead were often given attributes that defied the natural order of things. This suggests that humans conceptualize the Undead as possessing abnormal features to confirm their abnormal status. This differs from Options B and C, which suggest that the Undead are deified or that their natural attributes are rendered abnormal by changing their status. Option D is also inaccurate, as the passage does not mention that giving the Undead an abnormal status is a rejection of the natural order of things.

Hence, Option A is the correct answer.

VIDEO SOLUTION

10. A

Option A is a valid conclusion to draw from the lines - "*From out of the primal darkness of Mankind's earliest years, come whispers of eerie creatures, not quite alive (or alive in a way which we can understand), yet not quite dead either.*"

The statement mentions that in Mankind's earliest years, there were whispers of eerie creatures that were not quite alive or dead. This suggests that in these early years, there was a belief in the existence of such creatures. Option A accurately captures this idea. Option B is not a valid conclusion as the given statement does not mention anything about eerie creatures whispering about their own death. Option C is also incorrect since the passage does not mention that the creatures were only seen in the darkness. Similarly, Option D is not a valid conclusion as the given statement offers no information on how we can understand the lives of the eerie creatures.

Hence, Option A is the correct choice.

VIDEO SOLUTION

CAT Percentile Predictor



11. B

The passage underlines that the concept of the Undead, or creatures that are not quite alive or dead, has always been a part of human folklore. In ancient times, the Undead were ill-defined and vague, but as human societies became more sophisticated, the Undead took on more definite shapes and were often associated with supernatural powers, such as the ability to transform, drink blood, and influence human minds. The Undead have also been connected to the practice of magic, and in more recent times, specific names such as werewolf, vampire, and ghoul have become associated with the concept of the Undead. These names are often used to strike fear into the hearts of ordinary humans. The passage suggests that the Undead have evolved and developed over time, and as human societies have advanced, the Undead have become more defined and have gained more specific attributes. Overall, the passage discusses the long-standing presence of the Undead in human folklore and the evolution of their portrayal in various cultures. Option B aptly captures the above idea.

Option A is incorrect since the passage does not mention the transition from primitive thinking to the Age of Enlightenment. Option C is also inaccurate as the author does not emphasize the failure of human beings "to fully comprehend their environment" [not the focus]. Option D is not a complete description of the passage since it only mentions one aspect of the passage rather than the overall theme of the evolution of the concept of the Undead.

Hence, Option B is the correct choice.

VIDEO SOLUTION

12. B

The multiple negations indicate that if any of the given statements are false, they could be seen as being consistent with the information provided in the passage [i.e. they do not contradict the information provided in the passage]. Since the question involves "except," we need to find a valid statement based on the information provided [since if this statement is correct, it would not be consistent with the information in the passage].

We notice that Option B is consistent with the information provided in the passage: The passage states that as human societies became more sophisticated, the Undead took on more definite shapes and became more defined. It also mentions that from the Middle Ages and into the Age of Enlightenment, theories of the Undead continued to grow and develop. This suggests that the transition from the Middle Ages to the Age of Enlightenment saw new theories of the Undead. There will be inconsistencies if the statement in B is refuted or incorrect.

If the remaining options are false, they will support the discussion in the passage: Option A is not consistent with the passage as it states that the Undead remained vague and ill-defined, even as human societies strove to understand the horror they inspired, while the passage actually states that the Undead became more defined as human societies became more sophisticated. Option C is also incorrect because the passage does not mention that the growing sophistication of Mankind caused humans to stop believing in the Undead. Similarly, Option D is also inconsistent in its current form since the passage does not mention anything about the strength or stability of the relationship between Shamans and the Undead.

Hence, Option B is the correct choice.

 VIDEO SOLUTION

13. D

The fourth paragraph discusses the ways in which technology shapes society and the values and meanings that are inscribed in technological design. It suggests that technology, like laws and customs, represents the interests of those who use it and shapes the version of human nature that is sanctioned by society. The paragraph compares technology to laws and customs, stating that they are similar in the sense that they are institutions that shape the way people live. Therefore, the correct statement that captures the crux of the fourth paragraph is Option D [Technology, laws, and customs are not unlike each other if considered as institutions].

Option A: While it is true that technology, laws, and customs are comparable phenomena in this sense, the statement does not adequately capture the main point being made in the paragraph.

Option B: is incorrect because it misinterprets the main point of the fourth paragraph. While the paragraph does suggest that technology shapes the version of human nature that is sanctioned by society, it does not directly compare the effectiveness of technological environments and laws and customs in privileging certain dimensions of human nature.

Option C: is only partially true. While the fourth paragraph does mention the idea that technologies represent the interests of those who use them, it does not specifically state that automobiles represent the interest in mobility present in human nature. The paragraph mentions the automobile as an example of a technology that represents its users, but does not explicitly link it to the concept of mobility.

Hence, Option D is the correct choice.

 VIDEO SOLUTION

14. B

The passage specifically states that "critical theory of technology regards technologies as an environment rather than as a collection of tools," and that "technologies shape their inhabitants" in a way that is similar to laws and customs, which represent certain interests and values of those who live under their sway. There is no mention of the role of nature in shaping society or determining the limits of society's control over history, so options C and D cannot be supporting points. The statement in A wouldn't support the discussion in the passage because the author suggests that institutions such as laws and customs represent certain dimensions of human nature rather than human culture.

Hence, Option B is the correct choice.

 VIDEO SOLUTION

15. C

Option A: This statement is consistent with the arguments of the passage, which claim that the pattern of the capitalist rationalization of production arises from the problems of command over a disempowered and deskilled labor force and is present in many different contexts, including the factory and socialist systems.

Option B: This statement is consistent with the passage, which claims that Marx and Marcuse both contribute to the tradition of critical theory of technology, which seeks to understand the ways in which technological systems are shaped by and contribute to the reproduction of social and political hierarchies.

Option C: This statement contradicts an element discussed in the passage. The passage states that the pattern of the capitalist rationalization of production, which is marked by the centralization of power in institutions and organizations and the deskilling of the labor force, arises in many different contexts, including the factory, prisons, and the public sphere. It does not claim that the patterns in these different contexts are set by Foucault's prisons and Habermas' public sphere.

Option D: This statement is consistent with the passage, which claims that the pattern of the capitalist rationalization of production is present in many different contexts, including socialist systems.

Hence, Option C is the correct choice.

 VIDEO SOLUTION



16. D

Option A: The passage states that "laws of property represent the interest in ownership and control. Customs such as parental authority represent the interest of childhood in safety and growth." It then goes on to say that "interests such as these constitute the version of human nature sanctioned by society." This suggests that the concept of human nature is not fixed but rather emerges out of historical context and is shaped by society. Therefore, it can be inferred that the significance of parental authority to children's safety does not imply that parental authority is a permanent aspect of human nature.

Option B: This claim can be inferred - the author states that "critical theory of technology is a political theory of modernity with a normative dimension" and that it belongs to a tradition "according to which advances in the formal claims of human rights take centre stage while in the background centralization of ever more powerful public institutions and private organizations imposes an authoritarian social order." This suggests that the critical theory of technology argues



that as issues of human rights become more prominent, the social order becomes more authoritarian.

Option C: We are told that "a hermeneutics of technology must make explicit the meanings implicit in the devices we use and the rituals they script" and that "social histories of technologies such as the bicycle, artificial lighting or firearms have made important contributions to this type of analysis." This suggests that engaging with the social histories of technologies is necessary to understand their implicit and explicit meanings for us.

Option D: This cannot be inferred from the passage because the passage does not mention any costs or negative consequences of technologies privileging certain dimensions of human nature - it only discusses the idea that technologies represent the interests of their users and shape their behaviour and values; however, no claims are made about the potential negative impacts of this process.

Hence, Option D is the correct choice.

 VIDEO SOLUTION

17. D

The sentence would best fit in Blank 4 because it ties together the ideas presented in the paragraph. The paragraph states that people today have access to a vast amount of information, but it is not clear if they are more knowledgeable as a result. It also mentions that the readily available knowledge can lead students away from serious study and towards an inability to write effectively. The missing sentence introduces the idea that the Internet has both positive and negative effects on knowledge and learning. It fits well in Option 4 because it connects to the idea that the Internet has let people choose sources that reinforce their opinions rather than encouraging them to question inherited beliefs, which is the final point presented in the paragraph.

 VIDEO SOLUTION

18. C

The given passage discusses the role of petitioning in democratic governance; it highlights how petitioning can shape political agendas, recruit citizens to causes, give voice to the voteless, and apply the discipline of rhetorical argument to clarify points of view. The passage also compares petitioning to elections, stating that petitioning supplements the electoral process and enables ongoing engagement with the government. The author does not make a judgment about the relative superiority of petitioning or elections, but rather focuses on the specific ways in which petitioning can contribute to democratic functioning. Option C accurately summarizes the points discussed above.

Option A is incorrect because the passage does not state that petitioning is "an ideal form of representative democracy." It only mentions that petitioning can give voice to the voteless and apply the discipline of rhetorical argument, but it does not make a broader claim about the superiority of petitioning over other forms of democracy.

Option B is incorrect because the passage does not suggest that citizens are becoming less inclined to petitioning. In fact, the passage does not address the prevalence of petitioning in contemporary times at all.

Option D is incorrect because the passage does not state that petitioning is more representative of the collective voice than elections. It only mentions that petitioning can give voice to the voteless, but it does not make a comparison to elections in this regard. The passage also does not suggest that the functioning of democratic government would necessarily improve if we relied more on petitioning instead of holding periodic elections; it simply states that petitioning has been important to democratic functioning in the past, and that it supplements the electoral process by enabling ongoing engagement with the government.

 VIDEO SOLUTION

19. B

The sentence would best fit Blank 2 because it ties together the ideas presented in the paragraph. The paragraph describes the change in the mindset for wedding celebrations post covid. In the first two lines, the author mentions this mindset. The given sentence will fill the blank 2 as it restates the idea presented in the previous line that many couples are marrying secretly to ease the anxiety and pressure.

Thus, the correct option is B.

 VIDEO SOLUTION

20. C

The passage touches upon the historical relationship between Ethiopia and Europe during the medieval period. The traditional narrative of this relationship has portrayed Ethiopia as weak and in need of military assistance from Europe, but recent research has revealed that this narrative is not accurate and that Ethiopia was actually a well-connected and outward-looking culture that engaged in missions of diplomacy, faith, and commerce with Europe. The passage also notes that these new findings challenge historians to re-imagine the connections between Ethiopia and Europe during this period and to consider the role of Ethiopia as a more active participant in these relationships. Option C accurately reflects the main points made in the passage [that the traditional narrative of Ethiopia's relationship with Europe is inaccurate]

Option A is incorrect because it suggests that the connections between Ethiopia and Europe were primarily military and commercial in nature, which is not stated in the passage. Similarly, Option B implies that historians had a mistaken view of Ethiopia's relationship with Europe, which is not stated in the passage. The idea in Option D - which suggests that medieval texts were biased in favour of Europe and against Africa - cannot be understood from the discussion.

Hence, Option C is the correct choice.

 VIDEO SOLUTION



21. 2143

The given set of sentences underlines the relationship between commuting and productivity and how it can impact innovation. The idea is that shorter commutes can support innovation by giving employees more time in the office and opportunities for in-person collaboration, while longer commutes can be physically draining and can lead to decreased productivity. This correlation is first highlighted in statement 2, which sets the context for subsequent discussion. We note that statements 4 and 3 form a logical block since 4 presents an idea [work-from-home scenarios to improve productivity] and 3 extends on it [why the idea in 4 might not be effective]. The marker word 'some' in 4 helps us link it with statement 1, which discusses the opinions of "some company leaders." So 1 puts forth the belief of one group of company leaders, and 4 presents us with the opinion held by a few other company leaders. This allows us to form the logical block: [1-4-3]. We can place 2 at the beginning of this structure to obtain a coherent paragraph. Hence, the correct answer is 2143.

22. 3214

A brief reading of the sentences suggests that the paragraph is about the applications of the creative element in the world economy. Sentence 3 gives the introduction by contrasting the growth rate of the world economy and the creative industry. Statement 2 follows 3 by stating that the difference will be more if we consider the contribution of the creative element in other industries. Sentence 1 extends this idea by explaining how the creative element in product design can help sustain a competitive advantage. Sentence 4 concludes the paragraph by stating the implications of the application of the creative element mentioned in sentence 1.

Thus, the correct order will be 3-2-1-4.

23. D

The passage is about the limitations of our understanding of the universe and how it is connected. It suggests that our understanding of the distinction between animate and inanimate objects may be flawed because of our limited perspective, and that quantum mechanics reveals a different understanding of this distinction; it also suggests that there may be a pattern to the universe that we cannot fully grasp with our brains. Option D correctly captures these points.

Options A and B are incorrect because the passage does not state that the effect of stimuli is similar in inanimate objects compared to living beings or mention astigmatism or an erroneous view of reality. Option C is not understood since the passage does not suggest that inanimate objects are sentient or cognizant; it only highlights that the distinction between animate and inanimate objects disappears at the quantum scale.

Hence, Option D is the correct choice.

24. 2431

A brief reading of the sentences suggests that the paragraph is about the applications of a structural protein called Collagen. Sentence 2 introduces the paragraph by defining Collagen. 2-4 is an obvious link, as sentence 2 ends by stating that Collagen promotes skin regeneration, and sentence 4 explains why it is not widely used in the medical field despite this. Sentence 3 extends the idea given in sentence 4 by contrasting the bacteria affecting fish skins with human pathogens. Sentence 1 follows sentence 3 by providing the applications of fish skin collagen.

Thus, the correct order will be 2-4-3-1.

LRDI

**Explanation [25 - 29]:**

A total of 12 goals were scored in 8 matches and each player scored atleast one goal and no of goals scored by each one of them is distinct so the possible number of goals scored by the players can be (1,2,3,6) or (1,2,4,5).

From statement 4 we know that Bimal scored 4 goals and since Harita scored more goals than Bimal so we can say that Harita scored 5 goals and the only case possible for total goals scored by each of the players is (1,2,4,5).

Now using statement 1, statement 3 and statement 4 we can say that the three consecutive matches in which Bimal scored will be 5th, 6th and 7th matches as Harita scored in 4th and 8th matches and we get the following table:

| Matches | Goals Scored | Players  |
|---------|--------------|----------|
| Match-1 |              | Bimal-1  |
| Match-2 | 1            |          |
| Match-3 |              |          |
| Match-4 | 1            | Harita-1 |
| Match-5 |              | Bimal-1  |
| Match-6 | 1            | Bimal-1  |
| Match-7 |              | Bimal-1  |
| Match-8 | 1            | Harita-1 |

From statement 5 and 6, we can conclude that the highest number of goals were scored in Match 1.

Let the no. of goals scored in 3rd and 7th match be a each and no. of goals scored in 1st and 5th match be b and c respectively.

Therefore,  $2a+b+c=8$

If  $a=1$ , then  $b+c=6$  therefore possible solutions for b and c will be 2 and 4 only

If  $a=2$ , then  $b+c=4$  therefore possible solution for b and c will be 1 and 3 only but since highest goals scored is in Match 1 so then no. of goals scored in match 1 must be 3 and Harita must have scored 3 goals in match 1 as Harita scored 5 goals in exactly 3 matches. Therefore, we can see this is not possible because then the no. of goals scored in Match 1 becomes 4.

Therefore the only possible solution is  $a=1$ ,  $b=4$  and  $c=2$

| Matches | Goals Scored | Players           |
|---------|--------------|-------------------|
| Match-1 | 4            | Bimal-1, Harita-3 |
| Match-2 | 1            |                   |
| Match-3 | 1            |                   |
| Match-4 | 1            | Harita-1          |
| Match-5 | 2            | Bimal-1,          |
| Match-6 | 1            | Bimal-1           |
| Match-7 | 1            | Bimal-1           |
| Match-8 | 1            | Harita-1          |

The remaining 3 goals were scored in the match 2, 3 and 5 by Amla and Sarita in some order.

| Matches | Goals Scored | Players                |
|---------|--------------|------------------------|
| Match-1 | 4            | Bimal-1, Harita-3      |
| Match-2 | 1            | Amla/Sarita-1          |
| Match-3 | 1            | Amla/Sarita-1          |
| Match-4 | 1            | Harita-1               |
| Match-5 | 2            | Bimal-1, Amla/Sarita-1 |
| Match-6 | 1            | Bimal-1                |
| Match-7 | 1            | Bimal-1                |
| Match-8 | 1            | Harita-1               |

25. C

[VIDEO SOLUTION](#)



**Explanation [30 - 34]:**

No. of girls = 15

Let the no. of boys be  $x$

No. of singers = 6

No of boys who are singers = 4

Therefore, no of girls who are singers = 2

No of dancers = 10

No of boys who are dancers = 6

Therefore, no. of girls who are dancers = 4

No. of boys who are neither singers nor dancers =  $x-10$

No. of girls who are neither singers nor dancers = 9

|                                        | Not interested | 1-day event | 2-day event | 2-day event |
|----------------------------------------|----------------|-------------|-------------|-------------|
| <b>Boys (x)</b>                        |                |             |             |             |
| Singers (4)                            |                |             |             |             |
| Dancers (6)                            |                |             |             |             |
| Neither singers nor dancers ( $x-10$ ) |                |             |             |             |
| <b>Girls (15)</b>                      |                |             |             |             |
| Singers (2)                            |                |             |             |             |
| Dancers (4)                            |                |             |             |             |
| Neither singers nor dancers (9)        |                |             |             |             |

Now we fill the above table,

using statements 1 and 2, we get the following table

|                                    | Not interested | 1-day event | 2-day event | 3-day event |
|------------------------------------|----------------|-------------|-------------|-------------|
| <b>Boys (x)</b>                    | 0.2x           | 0.8x        | 0.6x        |             |
| Singers (4)                        |                |             |             |             |
| Dancers (6)                        |                |             |             |             |
| Neither singers nor dancers (x-10) |                |             |             |             |
| <b>Girls (15)</b>                  | 0              | 15          |             |             |
| Singers (2)                        | 0              | 2           |             |             |
| Dancers (4)                        | 0              | 4           |             |             |
| Neither singers nor dancers (9)    | 0              | 9           |             |             |

Let the number of girls who are interested in attending a 2-day event be a and the number of girls who are dancers and are interested in 2-day event be b.  
Now using statements 3 and 4, we get

|                                    | Not interested | 1-day event | 2-day event | 3-day event |
|------------------------------------|----------------|-------------|-------------|-------------|
| <b>Boys (x)</b>                    | 0.2x           | 0.8x        | 0.6x        |             |
| Singers (4)                        | 0              | 4           | 4           | 4           |
| Dancers (6)                        |                |             | $0.18x - 4$ | 2           |
| Neither singers nor dancers (x-10) |                |             | $0.42x$     |             |
| <b>Girls (15)</b>                  | 0              | 15          | a           | 0           |
| Singers (2)                        | 0              | 2           | $0.4a - b$  | 0           |
| Dancers (4)                        | 0              | 4           | b           | 0           |
| Neither singers nor dancers (9)    | 0              | 9           | $0.6a$      | 0           |

$$2 \leq 0.18x - 4 \leq 6$$

$$6 \leq 0.18x \leq 10$$

$0.18x$  should be integer for which x should be a multiple of 50, and  $0.18x$  lies between 6 and 10; therefore, the only possible value of x is 50.

|                                  | Not interested | 1-day event | 2-day event | 3-day event |
|----------------------------------|----------------|-------------|-------------|-------------|
| <b>Boys (50)</b>                 | 10             | 40          | 30          |             |
| Singers (4)                      | 0              | 4           | 4           | 4           |
| Dancers (6)                      |                |             | 5           | 2           |
| Neither singers nor dancers (40) |                |             | 21          |             |
| <b>Girls (15)</b>                | 0              | 15          | a           | 0           |
| Singers (2)                      | 0              | 2           | $0.4a - b$  | 0           |
| Dancers (4)                      | 0              | 4           | b           | 0           |
| Neither singers nor dancers (9)  | 0              | 9           | $0.6a$      | 0           |

From statement 5, we can say that,

$$4 + 0.4a - b = 5 + b + 1$$

$$\text{or, } 0.4a = 2 + 2b$$

$$\text{or, } a = 5(1 + b)$$

a should be a multiple of 5 as b is a whole number. So possible values of a can be 5, 10 or 15. Now, as the maximum value of b can be 4 and the maximum value of  $0.4a - b$  can be 2, so the only possible value of a satisfying the conditions above is 5. If  $a = 5$  then  $b = 1$ .



|                                  | Not interested | 1-day event | 2-day event | 3-day event |
|----------------------------------|----------------|-------------|-------------|-------------|
| <b>Boys (50)</b>                 | 10             | 40          | 30          |             |
| Singers (4)                      | 0              | 4           | 4           | 4           |
| Dancers (6)                      | 0/1            | 6/5         | 5           | 2           |
| Neither singers nor dancers (40) | 10/9           | 30/31       | 21          |             |
| <b>Girls (15)</b>                | 0              | 15          | 5           | 0           |
| Singers (2)                      | 0              | 2           | 2           | 0           |
| Dancers (4)                      | 0              | 4           | 0           | 0           |
| Neither singers nor dancers (9)  | 0              | 9           | 3           | 0           |

30.50

[VIDEO SOLUTION](#)

31.C

[VIDEO SOLUTION](#)

32.B

Total students interested in 2-day event = 30+5 = 35

Total students = 15 + 50 = 65

Fraction =  $\frac{35}{65} = \frac{7}{13}$

[VIDEO SOLUTION](#)

33.A

[VIDEO SOLUTION](#)

34.C

[VIDEO SOLUTION](#)



## CAT Daily Targets

Step by step, heading towards 99+ percentile



Explanation [35 - 39]:

| Amount(in 000s) | Tokens Received | Person |
|-----------------|-----------------|--------|
| 390             | 2,3,5,13        |        |
| 210             | 2,3,5,7         |        |
| 165             | 3,5,11          |        |
| 77              | 7,11            |        |
| 66              | 2,3,11          |        |

From statement 1 we can deduce that Fatima gave token 3 and Adhara gave token 13 and therefore from this we can say that Qahira received a funding of 77,000 and Pragnyaa received a funding of 390,000.

From statement 2, we know that Rashida received highest number of tokens and we already concluded that Pragnyaa received a funding of 390,000 so we can say that Rashida received a funding of 210,000.

Rashida did not received a token from Esther so we can also conclude that Esther gave the token number 11.

| Amount(in 000s) | Tokens Received | Person   |
|-----------------|-----------------|----------|
| 390             | 2,3,5,13        | Pragnyaa |
| 210             | 2,3,5,7         | Rashida  |
| 165             | 3,5,11          |          |
| 77              | 7,11            | Qahira   |
| 66              | 2,3,11          |          |

From statement 3 we can conclude that Dhanavi gave a token of 7, and Bethi Gave a token of either 2 or 5 and similarly Chhaya also gave a token of 2 or 5.

| Amount(in 000s) | Tokens Received | Person       |
|-----------------|-----------------|--------------|
| 390             | 2,3,5,13        | Pragnyaa     |
| 210             | 2,3,5,7         | Rashida      |
| 165             | 3,5,11          | Smera/Tantra |
| 77              | 7,11            | Qahira       |
| 66              | 2,3,11          | Smera/Tantra |

| Person  | Token |
|---------|-------|
| Adhara  | 13    |
| Bithi   | 2/5   |
| Chhaya  | 5/2   |
| Dhanavi | 7     |
| Esther  | 11    |
| Fathima | 3     |

**Alternate Explanation:**

$$390 = 2 \times 3 \times 5 \times 13$$

$$210 = 2 \times 3 \times 5 \times 7$$

$$165 = 3 \times 5 \times 11$$

$$77 = 7 \times 11$$

$$66 = 2 \times 3 \times 11$$

From the above information we can conclude that the number of times a particular token was given, therefore we get the following table,

| Token | No. of times given |
|-------|--------------------|
| 2     | 3                  |
| 3     | 4                  |
| 5     | 3                  |
| 7     | 2                  |
| 11    | 2                  |
| 13    | 1                  |

We know that there are five people who received the token and there are 6 people who awarded the token.

From, statement 1 we know that Fatima gave token to 4 people except Qahira so the token number given by Fatima is 3, and Adhara gave token only to Pragnyaa so the token number given by Adhara is 13. Therefore we can also say that Pragnyaa received 390,000 and Qahira received 77,000.

From statement 2, we know that Rashida received highest number of tokens and we already concluded that Pragnyaa received a funding of 390,000 so we can say that Rashida received a funding of 210,000. Rashida did not received a token from Esther so we can also conclude that Esther gave the token number 11.

|            | Pragnyaa<br>(390) | Qahira<br>(77) | Rashida<br>(210) | Smera | Tantra |
|------------|-------------------|----------------|------------------|-------|--------|
| Adhara(13) | ✓                 | X              | X                | X     | X      |
| Bithi      |                   |                | ✓                |       |        |
| Chhaya     |                   |                | ✓                |       |        |
| Dhanavi    |                   |                | ✓                |       |        |
| Esther(11) | X                 | ✓              | X                | ✓     | ✓      |
| Fathima(3) | ✓                 | X              | ✓                | ✓     | ✓      |

From statement 3, we can conclude that Dhanavi gave a token of 7, and Bethi Gave a token of either 2 or 5 and similarly Chhaya also gave a token of 2 or 5.

|              | Pragnyaa<br>(390) | Qahira<br>(77) | Rashida<br>(210) | Smera<br>(66/165) | Tantra<br>(165/66) |
|--------------|-------------------|----------------|------------------|-------------------|--------------------|
| Adhara (13)  | ✓                 | X              | X                | X                 | X                  |
| Bithi (2/5)  | ✓                 | X              | ✓                |                   |                    |
| Chhaya (2/5) | ✓                 | X              | ✓                |                   |                    |
| Dhanavi (7)  | X                 | ✓              | ✓                | X                 | X                  |
| Esther (11)  | X                 | ✓              | X                | ✓                 | ✓                  |
| Fathima (3)  | ✓                 | X              | ✓                | ✓                 | ✓                  |

35.2

Ans:2

[VIDEO SOLUTION](#)

36.A

Ans:A

[VIDEO SOLUTION](#)

37.3

Ans:3

[VIDEO SOLUTION](#)

38.3

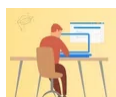
Ans:3

[VIDEO SOLUTION](#)

39.D

Ans:D

[VIDEO SOLUTION](#)



**FREE**

## CAT Sectional Tests

cracku

[Take Here](#)

40.A

In the east-west direction, a train starts from station M every 10 minutes.

Now the first train leaving station M is at 6:00 am.

Since every 10 minutes trains are available in east-west direction, the train timings leaving from M will be like 6:10 am, 6:20 am, 6:30 am and so on.

Given, Hari has reached station M by 8:05 am.

So the earliest by which Hari can catch a train from station M is 8:10 am.

Now there are 19 stations between M and n, out of which two stations are junctions.

Time taken to travel between two stations in the east-west direction is 2 minutes.

Therefore, the time for which the train was running between M and N (excluding the stoppage time) =  $20 \times 2 = 40$  minutes

Stoppage time at a junction is 2 minutes, while at the rest of the stations, it is 1 minute each.

Stoppage time for the train running between M and N =  $(17 \times 1) + (2 \times 2) = 21$  minutes

Therefore, total travel time =  $40 + 21 = 61$  minutes.

So the time by which Hari reaches N is 8:10 am + 61 minutes = 9:11 am

41. A

Priya can reach S from T via R or V.

**Case 1:- T-V-S**

In the east-west direction, the first train from P arrives at T at time = 6 am +  $(4 \times 2) + (3 \times 1) = 11$  minutes = 6:11 am

Since T is a junction so this train will halt for 2 minutes at T and leave at 6:13.

Priya boards a east-west train then Priya will board a train for V from T at 10:33 am.

There are 9 stations between T and V

Travelling time between T and V =  $(10 \times 2) + (9 \times 1) = 29$  minutes

Therefore, Priya will reach V latest by 10:33 am + 29 minutes = 11:02 am

In the north-south direction, the first train from D arrives at V at time = 6 am +  $(3 \times 3) + (2 \times 1) = 11$  minutes = 6:11 am

Since V is a junction so this train will halt for 2 minutes at V and leave at 6:13.

Since every 15 minutes, a train starts from D in the north-south direction, so the latest by which Priya will be able to board such a train from V is at 11:13 am.

There are 3 stations between V and S

Travelling time between R and S =  $(4 \times 3) + ((3 \times 1)) = 15$  minutes

Time by which she reaches S = 11:13 + 15 minutes = 11:28 am

**Case 2:- T-R-S**

In the north-south direction, the first train from B arrives at T at time = 6 am +  $(3 \times 3) + (2 \times 1) = 11$  minutes = 6:11 am

Since T is a junction so this train will halt for 2 minutes at T and leave at 6:13.

Since every 15 minutes a train starts from P in the east-west direction so the latest by which Priya will be able to board such a train is at 10:28 am.

Now since she will be able to board a north-south train earlier than the east-west train so Priya will board a train for R from T at 10:28 am.

There are 3 stations between T and R

Travelling time between T and R =  $(4 \times 3) + ((3 \times 1)) = 15$  minutes

Therefore, Priya will reach R latest by 10:43 am

In the east-west direction, the first train from M arrives at R at time = 6 am +  $(4 \times 2) + (3 \times 1) = 11$  minutes = 6:11 am

Since R is a junction so this train will halt for 2 minutes at R and leave at 6:13.

Since every 10 minutes, a train starts from M in the east-west direction, so the latest by which Priya will be able to board such a train is at 10:43 am.

There are 9 stations between R and S

Travelling time between R and S =  $(10 \times 2) + (9 \times 1) = 29$  minutes

Time by which she reaches S = 10:43 + 29 minutes = 11:12 am

We are getting shorter time in case 2. So 11:12 am is the answer

42. A

Travelling time between S and R =  $(10 \times 2) + (9 \times 1) = 29$  minutes

There is a stoppage of 2 minutes at R

Travelling time between R and B =  $(7 \times 3) + (1 \times 2) + (5 \times 1) = 28$  minutes

In the north-south direction, the first train from A arrives at R at time = 6 am +  $(3 \times 3) + (2 \times 1) = 6:11$  am.

Since R is a junction so this train will halt for 2 minutes at R and leave at 6:13.

Every 15 minutes, a train starts from A in the north-south direction.

The last train that leaves A will be at 12:00 am and it will leave R at 12:13 am, so Haripriya must reach R till 12:13 am.

Travelling time between S and R =  $(10 \times 2) + (9 \times 1) = 29$  minutes

So Haripriya must board the train at S by 11:44 pm

In the east-west direction, the first train from N arrives at S at time = 6 am +  $(6 \times 2) + (5 \times 1) = 6:17$  am.

Since S is a junction so this train will halt for 2 minutes at S and leave at 6:19.

Every 10 minutes, a train starts from N in the east-west direction.

Therefore, Haripriya should board the train which leaves S at 11:39.

43. 8

Travel time between A and B =  $(10 \times 3) + (7 \times 1) + (2 \times 2) = 41$  minutes

After completing a journey, a train must rest for 15 minutes at least before starting again.

So if a train starts from 6 am from A to B, then the latest by which that train will start from B to A will be at 7 am, as in the north-south direction, a train starts from A and B every 15 minutes.

So the total no. of trains required =  $\left(\frac{60}{15}\right) \times 2 = 8$

VIDEO SOLUTION

44.48

Travel time between A and B =  $(10 \times 3) + (7 \times 1) + (2 \times 2) = 41$  minutes

After completing a journey, a train must rest for 15 minutes at least before starting again.

So if a train starts from 6 am from A to B, then the latest by which that train will start from B to A will be at 7 am, as in the north-south direction, a train starts from A and B every 15 minutes.

So the total no. of trains required for the north-south lines =  $\left(\frac{60}{15}\right) \times 2 \times 2 = 16$

Travel time between M and N =  $(20 \times 2) + (17 \times 1) + (2 \times 2) = 61$  minutes

After completing a journey, a train must rest for 15 minutes at least before starting again.

So if a train starts from 6 am from M to N, then the latest by which that train will start from N to M will be at 7:20 am, as in the east-west direction, a train starts from M and N every 15 minutes.

So the total no. of trains required for the east-west lines =  $\left(\frac{80}{10}\right) \times 2 \times 2 = 32$

Total no. of trains required to service the city =  $16 + 32 = 48$

VIDEO SOLUTION



Quant

45.43200

In the question, it is given that the ratio of number of literate males to literate females is 2 : 3.

Given, the number of literate males = 3600

The number of literate females =  $\frac{3600}{2} \times 3 = 5400$

Males : Females = 5 : 4

Let the number of males be  $5y$  and the number of females be  $4y$

Illiterate males =  $5y - 3600$

Illiterate females =  $4y - 5400$

It is given,

$$\frac{5y-3600}{4y-5400} = \frac{4}{3}$$

$$15y - 10800 = 16y - 21600$$

$$y = 10800$$

$$\text{Number of females} = 4y = 4 \times 10800 = 43200$$

VIDEO SOLUTION

46. C

Let the original number of students be 'n' whose average weight is 'x'

Let the number of students added be 'm' and the average weight will be  $x + 3$

We need to find the value of  $n : m$

It is given, average weight of students in a class increased by 0.6 after new students are added.

Therefore,

$$\frac{nx+m(x+3)}{n+m} = x + 0.6$$

$$nx + mx + 3m = mx + nx + 0.6n + 0.6m$$

$$2.4m = 0.6n$$

$$4m = n$$

$$\frac{n}{m} = \frac{4}{1}$$

The answer is option C.

VIDEO SOLUTION

47. C

It is given,

$$S_n = 2n^2 + n$$

$$S_{n-1} = 2(n-1)^2 + (n-1)$$

$$S_{n-1} = 2n^2 - 3n + 1$$

$$T_n = S_n - S_{n-1} = 2n^2 + n - 2n^2 + 3n - 1 = 4n - 1$$

$$T_n = 4n - 1$$

The terms are 3, 7, 11, 15, 19, 23, 27,.....

27 is the first term in the series divisible by 9.

27 is the 7th term.

Therefore, the least possible value of n is 7.

The answer is option C.

[VIDEO SOLUTION](#)

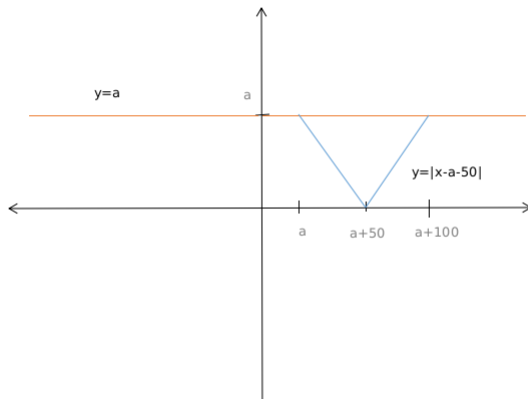
48. C

$x > a$ , so  $|x-a| = x-a$

$x < 100$ , so  $|x-100| = 100-x$

$$f(x) = (x-a) + (100-x) + |x-a-50| = 100$$

$$\text{or, } |x-a-50| = a$$



From the graph we can see that when  $x=a$  then

$$|x-a-50|=a$$

$$\text{or, } a = 50$$

Similarly when  $x=a+100$

$$|x-a-50|=a$$

$$\text{or, } a = 50$$

So value of a is 50 when  $f(x)$  is 100.

[VIDEO SOLUTION](#)

49. B



M - First meeting point

Let the speeds of trains A and B be 'a' and 'b', respectively.

$$\frac{x}{a} = \frac{D-x}{b}$$

It is given,

$$\frac{D}{a} = 10 \text{ and } \frac{x}{b} = 9$$

$$\frac{x}{D} = \frac{D-x}{9}$$

$$\frac{10x}{D} = \frac{9D-9x}{x}$$

$$10x^2 = 9D^2 - 9Dx$$

$$10x^2 + 9Dx - 9D^2 = 0$$

$$\text{Solving, we get } x = \frac{3D}{5}$$

$$\frac{x}{b} = 9$$

$$\frac{3D}{b \times 5} = 9$$

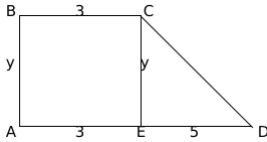
$$\frac{D}{b} = 15$$

The total time taken by train B to travel from station Y to station X is 15 minutes.

The answer is option B



50.66



$$CD = \sqrt{y^2 + 25}$$

$$11 + y + \sqrt{y^2 + 25} = 36$$

$$\sqrt{y^2 + 25} = 25 - y$$

$$y^2 + 25 = 25^2 + y^2 - 50y$$

$$2y = 24$$

$$y = 12$$

$$\text{Area of trapezium} = 3y + \frac{5y}{2} = \frac{11y}{2} = \frac{11}{2} (12) = 66$$

51. A

It is given,

$7C = 30P = 9A$  and Ankita bought 4C, 14P and 6A.

Let  $7C = 30P = 9A = 630k$

$C = 90k$ ,  $P = 21k$ , and  $A = 70k$

Cost price of 4C, 14P and 6A =  $4(90k) + 14(21k) + 6(70k) = 1074k$

Marked up price =  $1074k + 1752$

$$S.P = \frac{1}{6} (1074k + 1752) + \left(\frac{4}{5}\right) \left(\frac{5}{6}\right) (1074k + 1752) = \frac{5}{6} (1074k + 1752)$$

$S.P - C.P = \text{profit}$

$$1460 - \frac{1074k}{6} = 744$$

$$\frac{1074k}{6} = 716$$

$$k = 4$$

Money spent on buying almonds =  $420k = 420 \times 4 = \text{Rs } 1680$

The answer is option A.

52. 82

A is the HCF of  $3^k + 4^k + 5^k$  for different values of k.

For  $k = 1$ , value is 12

For  $k = 2$ , value is 50

For  $k = 3$ , value is 216

HCF is 2. Therefore,  $A = 2$

$$4^k + 3(4^k) + 4^{k+2} = 4^k(1 + 3) + 4^{k+2} = 4^{k+1} + 4^{k+2} = 4^{k+1}(1 + 4) = 5 \cdot 4^{k+1}$$

HCF of the values is when  $k = 1$ , i.e.  $5 \cdot 16 = 80$

Therefore,  $B = 80$

$$A + B = 82$$

53. C

$b^2 < 4ac$  means that the discriminant is less than 0. Therefore,  $f(x) > 0$  for all x if the coefficient of  $x^2$  is positive, and  $f(x) < 0$  for all x if the coefficient of  $x^2$  is negative.

We are given that  $f(m) < 0$  and m is an integer.

So the set containing values of m will either be empty if the coefficient of  $x^2$  is positive, or it will be a set of all integers if the coefficient of  $x^2$  is negative.

54. 84

Let the number of balloons each child received be 2a, 2b, 2c and 2d

$$2a + 2b + 2c + 2d = 20$$

$$a + b + c + d = 10$$

Each of them should get more than zero balloons.

$$\text{Therefore, total number of ways} = (n-1)_{C_{r-1}} = (10-1)_{C_{4-1}} = 9C_3 = 84$$

[VIDEO SOLUTION](#)

## 100+ CAT Quant Questions



55. **A**

$$a(a+b+1) = 14 \dots\dots (1)$$

$$b(a+b+1) = 28 \dots\dots (2)$$

$$\frac{a}{b} = \frac{1}{2}$$

$$b = 2a$$

Substituting in (1), we get

$$a(3a+1) = 14$$

$$3a^2 + a - 14 = 0$$

$$3a^2 - 6a + 7a - 14 = 0$$

$$3a(a-2) + 7(a-2) = 0$$

Given, a and b are natural numbers.

Therefore, a = 2 and b = 4

$$2a + b = 2(2) + 4 = 8$$

The answer is option A.

[VIDEO SOLUTION](#)

56. **160**

Total syrup - 110 kg

Total juice - 120 kg

It is given, cost price of syrup is 20% less than the cost price of juice.

Let the cost price of juice per kg be 10CP

Cost price of syrup per kg is 8CP

10kg syrup → cost price = 80CP

It is given, 10kg syrup is sold at 10% profit. This implies selling price =  $1.1 \times 80\text{CP} = 88\text{CP}$

20kg juice → cost price = 200CP

It is given, 20kg juice is sold at 20% profit. This implies selling price =  $1.2 \times 200\text{CP} = 240\text{CP}$

It is given, Mixing the remaining juice and syrup, Amal sells the mixture at ₹ 308.32 per kg

Selling price of the remaining mixture =  $308.32 \times 200 = \text{Rs } 61664$

Total S.P =  $61664 + 328\text{CP}$

Total C.P =  $880\text{CP} + 1200\text{CP} = 2080\text{CP}$

Overall profit = 64%

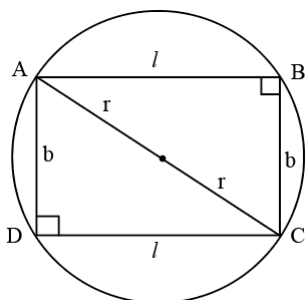
$$61664 + 328\text{CP} = \frac{164}{100} (2080\text{CP})$$

Solving, we get CP = 20

Cost price for syrup per kg =  $8\text{CP} = 8 \times 20 = \text{Rs } 160$

[VIDEO SOLUTION](#)

57. **B**



$$A = lb$$

$$l^2 + b^2 = 4r^2$$

$$P = 2(l + b)$$

$$\frac{P}{2} = l + b$$

Squaring on both the sides, we get

$$\frac{P^2}{4} = l^2 + b^2 + 2lb$$

$$\frac{P^2}{4} = 4r^2 + 2lb$$

$$\frac{P^2}{8} - 2r^2 = lb$$

The answer is option B.

[VIDEO SOLUTION](#)

58. C

It is given that average of three numbers is 13.

$$\text{Sum} = 3 \times 13 = 39$$

It is given,  $\frac{39+n}{4}$  is a odd number.

Minimum value  $\frac{39+n}{4}$  can take such that n is a natural number is 11

$$\frac{39+n}{4} = 11$$

$$n = 5$$

The answer is option C.

[VIDEO SOLUTION](#)

59. D

Lemon juice : sugar syrup in the mixture is 1:1, i.e. 50% Lemon juice and 50% sugar syrup.

In sugar syrup, 100% is sugar syrup.

These two are mixed in the ratio 1:3.

$$\text{Lemon juice} = \frac{1(50\%)}{1+3}$$

$$\text{Sugar syrup} = \frac{1(50\%) + 3(100\%)}{1+3} = \frac{350}{4}$$

$$\text{Required ratio} = 50:350 = 1:7$$

The answer is option D.

[VIDEO SOLUTION](#)

## 100+ CAT DILR Questions



60. C

In the question, it is given that the equation  $|x+a| + |x-1| = 2$  has an infinite number of solutions for any value of x. This is possible when x in  $|x+a|$  and x in  $|x-1|$  cancels out.

Case 1:

$$x+a < 0, x-1 \geq 0$$

$$-a-x+x-1=2$$

$$a=-3$$

Case 2:

$$x+a \geq 0 \text{ and } x-1 < 0$$

$$x+a-x+1=2$$

$$a=1$$

Largest value of a is 1.

The answer is option C.

[VIDEO SOLUTION](#)

61. A

Let n, s, d and t be the number of students who likes none of the drinks, exactly one drink, exactly 2 drinks and all three drinks, respectively.

It is given,

$$n+s+d+t=100 \dots\dots (1)$$

$$s+2d+3t=73+80+52$$

$$s+2d+3t=205 \dots\dots (2)$$

(2)-(1), we get

$$d+2t-n=105$$

Maximum value t can take is 52, i.e. t = 52, d = 1 and n = 0

Minimum value t can take is 5, i.e. t = 5, d = 95 and n = 0 (This also satisfies equation (1))

Difference =  $52 - 5 = 47$

The answer is option A.

[VIDEO SOLUTION](#)

62. **D**

In a parallelogram, two diagonals of parallelogram bisect each other, which concludes that mid-point of both diagonals are the same.

Midpoint of AC =  $\left(\frac{1-2}{2}, \frac{1+8}{2}\right)$

Let the coordinates of vertex D be (x,y)

$\left(\frac{x+3}{2}, \frac{y+4}{2}\right) = \left(\frac{1-2}{2}, \frac{1+8}{2}\right)$

$x = -4$  and  $y = 5$

The answer is option D.

[VIDEO SOLUTION](#)

63. **34**

It is given,  $y(x+z) = 19$

y cannot be 19.

If  $y = 19$ ,  $x+z = 1$  which is not possible when both x and z are natural numbers.

Therefore,  $y = 1$  and  $x+z = 19$

It is given,  $z(x+y) = 51$

z can take values 3 and 17

Case 1:

If  $z = 3$ ,  $y = 1$  and  $x = 16$

$xyz = 3 \times 1 \times 16 = 48$

Case 2:

If  $z = 17$ ,  $y = 1$  and  $x = 2$

$xyz = 17 \times 1 \times 2 = 34$

Minimum value xyz can take is 34.

[VIDEO SOLUTION](#)

64. **A**

Let the savings invested in first part and second part be 'x' and 'y', respectively.

It is given,

$$\frac{x \times 15 \times 4}{100} = \frac{y \times 12 \times 3}{100}$$

$$60x = 36y$$

$$5x = 3y$$

$$\text{Required percentage} = \frac{x}{x+y} \times 100 = \frac{3}{3+5} \times 100 = 37.5\%$$

The answer is option A.

[VIDEO SOLUTION](#)

## 100+ CAT VARC Questions



65. **111**

Let the number of persons standing ahead and behind of Pinky be 3a and 5a.

Total number of persons =  $3a + 5a + 1$  (including pinky) =  $8a + 1$

$$8a + 1 < 300$$

$$8a < 299$$

$$a < 37.375$$

Maximum value a can take is 37.

The maximum possible number of persons standing ahead of Pinky =  $3a = 3 \times 37 = 111$

[VIDEO SOLUTION](#)

66. **44**

It is given,

$$\sum_{n=1}^N \left[ \frac{1}{5} + \frac{n}{25} \right] = 25$$

$$\sum_{n=1}^N \left[ \frac{5+n}{25} \right] = 25$$

For  $n = 1$  to  $n = 19$ , value of function is zero.

For  $n = 20$  to  $n = 44$ , value of function will be 1.

$$44 = 20 + n - 1$$

$n = 25$  which is equal to given value.

This implies  $N = 44$

 VIDEO SOLUTION